

Cramer's rule

- If the determinant of a 3×3 matrix **A** is non-zero, we can solve the linear system **A** **x** = **b** by Cramer's rule.
- The solution to

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \quad \text{or equivalently} \quad \underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}}_{\mathbf{b}}$$

is

$$x_1 = \frac{1}{\det \mathbf{A}} \begin{vmatrix} \color{red}{b_1} & a_{12} & a_{13} \\ \color{red}{b_2} & a_{22} & a_{23} \\ \color{red}{b_3} & a_{32} & a_{33} \end{vmatrix} \quad x_2 = \frac{1}{\det \mathbf{A}} \begin{vmatrix} a_{11} & \color{red}{b_1} & a_{13} \\ a_{21} & \color{red}{b_2} & a_{23} \\ a_{31} & \color{red}{b_3} & a_{33} \end{vmatrix} \quad x_3 = \frac{1}{\det \mathbf{A}} \begin{vmatrix} a_{11} & a_{12} & \color{red}{b_1} \\ a_{21} & a_{22} & \color{red}{b_2} \\ a_{31} & a_{32} & \color{red}{b_3} \end{vmatrix}$$